DATA ENGINEERING

Dimensional Modeling Visual Guide

Facts, dimensions, grain, and SCDs.
Design tables humans can actually query.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Facts vs Dimensions

The two table types everything else is built from.

Fact table

- Long and narrow: billions of rows
- Numeric measures: amount, qty
- Foreign keys to dimensions

Dimension table

- Short and wide: thousands of rows
- Text attributes: name, category
- Surrogate primary keys

Facts answer 'how much / how many'; dimensions answer 'by what'. Keep measures additive and dimensions denormalized.

STEP 2/6

Grain: The Most Important Decision

Grain = what one row represents. Declare it before modeling anything.

"One row per order line item"

the grain statement

Too coarse (one row per order): you can't analyze products.

Too fine (one row per scan event): billions of rows nobody queries.

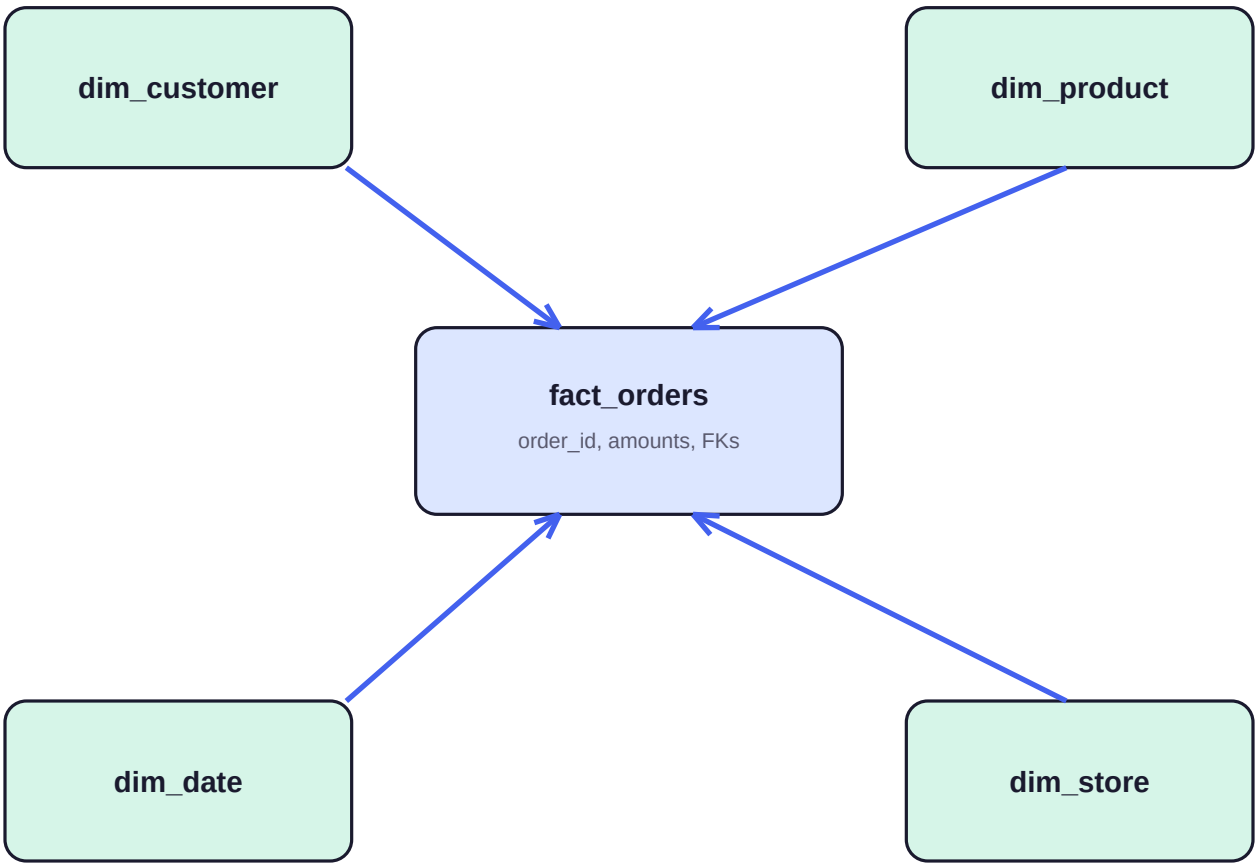
Mixed grain in one fact: measures stop being comparable — never do this.

If you can't state the grain in one sentence, you don't understand the business process yet. Go back and ask.

STEP 3/6

The Star: Fact in the Center

One fact table joined directly to denormalized dimensions.



Single-hop joins make star schemas fast and obvious — BI tools generate correct SQL from them with minimal configuration.

STEP 4/6

Slowly Changing Dimensions

Customers move, products get rebranded. How do you keep history?

Type 1
overwrite — no history

Type 2
new row — full history

Type 3
new column — limited history

Type 1 for corrections ('the name was misspelled'). Type 2 for real history ('she moved from FL to MA'). Type 3 when you only need previous vs current.

STEP 5/6

Type 2 in Detail

New row per change, with effective dates and a current flag.

```
sk   | cust_id | city        | eff_from   | eff_to     | current
101  | C-42    | Orlando     | 2024-01-01 | 2026-09-30 | N
102  | C-42    | Boston      | 2026-10-01 | 9999-12-31 | Y
```

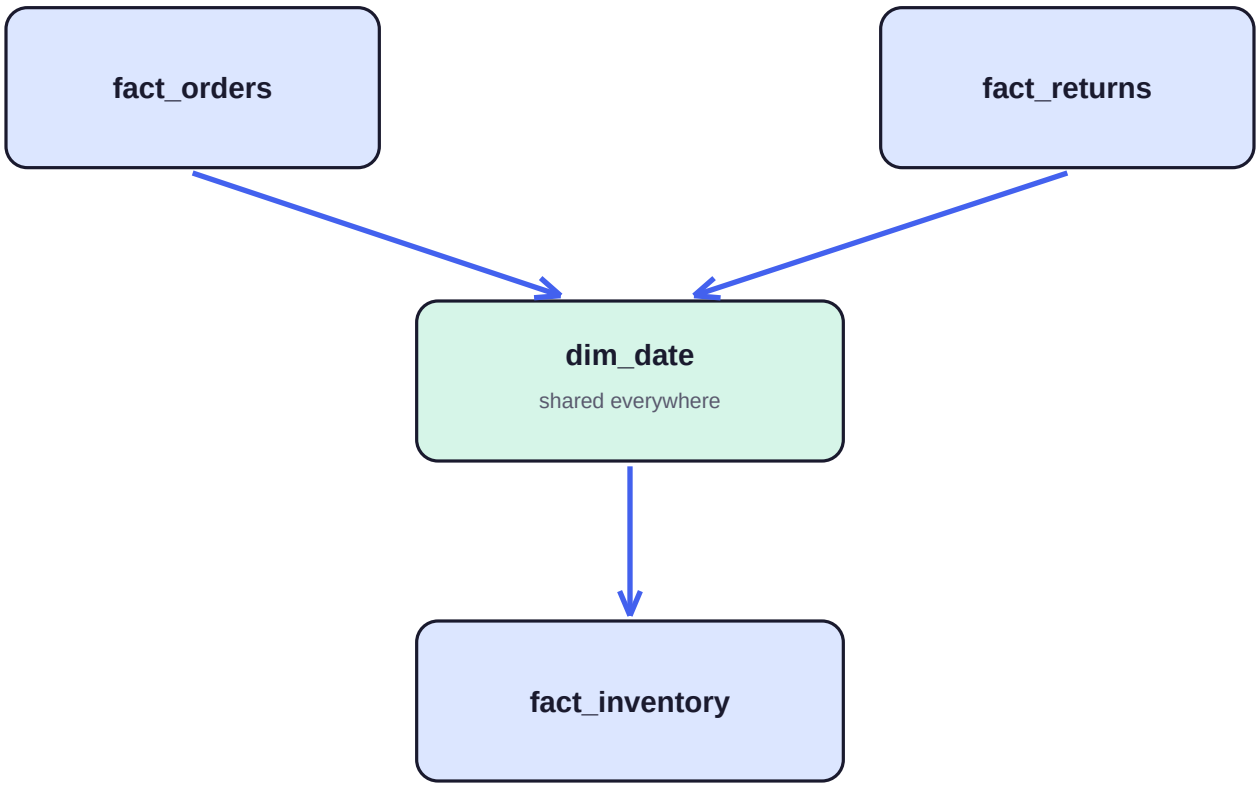
Surrogate key (sk) is the dimension's PK; the natural key (cust_id) repeats. Facts join on sk to pin the historically-correct version.

- dbt snapshots automate Type 2 with timestamp or check strategies
- Current flag + eff_to = 9999-12-31 makes 'as of now' queries trivial
- Point-in-time joins: fact date BETWEEN eff_from AND eff_to

STEP 6/6

Conformed Dimensions

One dim_date shared by every fact = consistent reporting.



Conformed dimensions are the enterprise bus: 'month' means the same in sales, finance, and marketing. Build them once, reuse forever.

STEP ✓/✓

Key Takeaways

Dimensional Modeling — Remember

- Facts hold measurable events; dimensions hold descriptive context. Facts are long, dimensions are wide.
- Declare the grain in one sentence before modeling — mixed grain breaks comparability.
- Star schemas join facts directly to denormalized dimensions: fast, simple, BI-friendly.
- SCD Type 1 overwrites, Type 2 versions history with effective dates, Type 3 keeps previous values.
- Conformed dimensions (one dim_date for all facts) keep the whole warehouse speaking one language.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev